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Application Summary

Competition Details

Competition Title:	Digital Transformation Research Corps
Category:	Undergraduate and Graduate Research Opportunity
Cycle:	AY2025-2026
Submission Deadline:	01/22/2026 11:59 PM

Application Information

Application Title:	TWAIN: The WashU AI-Assisted Platform for Simulation From Narration
Application ID:	1908
Submission Date:	01/8/2026 11:21 PM

Personal Details

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PI Availability:	Available to meet weekly throughout the program
Project Title:	TWAIN: The WashU AI-Assisted Platform for Simulation From Narration
Project Abstract:	TWAIN (The WashU AI-Assisted Platform for Simulation From Narration) seeks to reduce the manual work required to turn a scientific question into a simulation study. The project will deliver a prototype that takes a researcher's text or spoken description, uses an AI language model to identify intent, and asks targeted follow-up questions when details are missing. It then produces a structured, validated workflow that can run on machines or computing clusters. TWAIN will separate user-stated constraints from system-selected defaults, automatically prepare and check setup files, launch and monitor runs, and connect analyses and visual summaries to a record of decisions and assumptions. The platform will support both direct "run this simulation" requests and broader questions about properties by proposing alternative pathways and helping users weigh

cost-accuracy trade-offs. By standardizing workflow capture and enabling cross-device review and monitoring, TWAIN is expected to improve productivity, reproducibility, and training across computational research communities.

What external sponsors are being targeted upon completion of this 10-week program:	National Science Foundation, Office of Advanced Cyberinfrastructure — Pathways to Enable Open-Source Ecosystems (https://www.nsf.gov/funding/opportunities/pose-pathways-enable-open-source-ecosystems) U.S. Department of Energy, Office of Science — Advanced Scientific Computing Research program (https://science.osti.gov/ascr/-/media/grants/pdf/foas/2026/DE-FOA-0003600-000001.pdf) National Institutes of Health — Research Software Engineer Award (https://simpler.grants.gov/opportunity/002d36ed-9901-45b6-a1c4-2642dbf64146)
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TWAIN: The WashU AI-Assisted Platform for Simulation From Narration

1. Project Summary and Research Objectives

Computational research workflows in chemistry, physics, materials science, and biophysics still require substantial manual effort to translate a scientific goal into validated simulation inputs, orchestrate execution on local/HPC resources, and interpret results. This project will deliver **TWAIN** (*The WashU AI-Assisted Platform for Simulation From Narration*), a minimum viable product (MVP) that converts researcher narratives (text or speech) into structured, validated, and executable scientific workflows with full provenance. The core research objective is a typed **intent specification** layer (JSON/YAML) that supports both (1) **simulation-explicit requests** and (2) **property-driven questions**, where the system proposes, justifies, and iteratively refines candidate simulation pathways.



Figure 1. **TWAIN** narrative-to-workflow pipeline combining semantic intent parsing, interactive decision support, automated execution, and post-processing.

2. Project Overview and Digital Transformation Impact

Digital transformation impact. **TWAIN** shifts simulation setup and execution from a script-and-file-centric process to a narrative-and-decision-centric workflow. By interpreting a researcher's intent, asking targeted clarifying questions, and generating machine-readable specifications that drive automated input generation, job submission, logging, and post-processing, the system lowers barriers to advanced computation and improves reproducibility and productivity.

Advancing research and infrastructure. The MVP standardizes workflow capture, validation, and provenance across heterogeneous codes and computing environments. It also functions as an educational decision-support tool by making modeling assumptions explicit, reviewable, and shareable.

Cross-device relevance. The MVP demonstrates device-appropriate roles across **macOS** (workflow review and analysis), **iOS/iPadOS** (narrative capture and interactive refinement), **visionOS** (immersive inspection of structures, trajectories, and workflow branches), and **watchOS** (status updates and approval prompts), enabling continuous engagement while keeping an authoritative workflow state synchronized across devices.

Cross-device loop. The MVP supports a coordinated loop in which intent is captured and refined on mobile; workflows are reviewed, executed, and analyzed on desktop; and execution is monitored, and approvals are provided via lightweight watch prompts, with optional immersive inspection on visionOS.

3. Innovation Compared to Current Practice

This approach departs from current practice in three key ways. First, it introduces a typed intent representation that bridges narrative input to executable workflows, explicitly separating user-provided constraints from system-inferred choices and validated defaults. Second, it adds interactive decision support for property-driven questions by proposing multiple candidate modeling pathways, clearly articulating cost–fidelity trade-offs, and converging on an appropriate approach through targeted clarification. Third, it enables cross-device, stateful workflow interaction, supporting a coordinated “phone → desktop → watch” loop in which researchers can capture intent, review and refine workflows, monitor execution, and provide iterative approvals as work progresses.

4. Technical Approach

The **TWAIN** MVP will be implemented as a set of modular services unified by a typed intent specification and a shared provenance model. First, an LLM-assisted semantic parsing component converts researcher narratives into a typed workflow specification and detects ambiguity, prompting the user with targeted clarifying questions when needed. For property-driven requests, a decision-support module then proposes candidate modeling strategies, explains their assumptions, and surfaces cost–fidelity trade-offs so the workflow can converge on an appropriate pathway. Once an approach is selected, an input-generation and validation service produces representative simulation inputs—such as VASP-style DFT inputs, structures constructed via pymatgen/ASE/FreeBird.jl, or biomolecular configurations—while running automated consistency checks to catch common errors early. The platform then orchestrates execution on local machines and HPC systems via scheduler scripts, capturing synchronized logs, metadata, and end-to-end run provenance that remain consistent across devices. Finally, post-processing and visualization modules apply templated analyses to extract key quantities and generate publication-ready plots and dashboards that connect results back to the original intent and recorded assumptions.

5. Proposed Activities, Phases, Milestones, and Deliverables

The team will execute a 10-week plan to deliver a working, documented prototype validated on three representative narrative workflows spanning chemistry, physics, materials, and biophysics.

Weeks	Milestones	Deliverables
1–2	Requirements & schema	Schema (JSON/YAML); use cases; MVP metrics
3–4	Parsing & refinement	Parsing module; clarification and validation tests
5–6	Input generation	Input generators, validators, and samples
7–8	Orchestration & provenance	Local/HPC scripts; logging and provenance capture
9–10	Integration & visualization	TWAIN MVP; plots/dashboards; demo + guide

Anticipated program-period outputs:

- Documented, integrated **TWAIN** codebase (schema, parsing, orchestration, analytics).
- Three validated, end-to-end narrative workflows spanning distinct domains.
- Cross-device **TWAIN** MVP demonstration, plus demo video(s) and a user guide.

- External proposal assets (technical narrative, architecture, preliminary evaluation).

6. Anticipated Technical Needs

The MVP will require a modest but targeted technical environment. Development will be carried out on macOS machines, with access to a campus HPC system to support small validation runs and integration with common schedulers such as Slurm or PBS. The software stack will center on Python and Julia workflow and analysis libraries—including tools such as pymatgen, ASE, and FreeBird.jl—paired with SwiftUI to implement the cross-device interfaces. To enable rapid iteration and robust regression testing, the project will use benchmark structures and precomputed outputs; no sensitive datasets are needed for the MVP. In addition, the system will depend on LLM access for schema-constrained intent parsing and dialogue, along with prompt and version tracking to support reproducibility and evaluation.

7. Student Technical and Experiential Learning

Student collaborators (1–4) will work in cross-functional roles across the MVP stack and will gain hands-on experience in four core areas. They will develop skills in human–AI interaction by designing and testing clarifying-question strategies, analyzing failure modes, and implementing schema- and validation-based guardrails to improve reliability. They will also strengthen research software engineering practice through modular system design, automated testing and continuous integration, documentation, and packaging approaches that support reuse and long-term maintainability. In addition, students will learn practical HPC workflow and reproducibility techniques, including building job-submission templates, monitoring runs, capturing provenance, and producing reproducible analysis artifacts. Finally, they will gain cross-platform development experience by implementing a minimal SwiftUI interface and a shared state model that helps translate a research prototype into a usable, device-integrated tool.

8. External Funding Competitiveness and Sustainability

DTRC participation will generate MVP evidence—working prototype, evaluation metrics, and cross-domain workflows—needed to compete for external support from agencies that fund AI-enabled scientific discovery and cyberinfrastructure (e.g., NSF/NIH; DOE Office of Science). Sustainability will be pursued through open, documented software releases; integration with campus research computing support pathways; and submission of at least one external proposal within 6–12 months, leveraging MVP results and DI2 proposal development support.

9. Project Readiness, Feasibility, and Risk Management

The project is at a concept-to-prototype stage, with defined use cases, an initial architecture, and high-level interface and workflow designs. Key risks have been identified along with concrete mitigations. To address LLM reliability, the system will use schema-constrained outputs and validation gates, require explicit user confirmation before execution, and maintain regression tests based on representative narratives to detect drift and failure modes early. To manage HPC variability, the MVP will target a single scheduler environment first and rely on a well-defined set of module and container assumptions, expanding to additional scheduler adapters only after core functionality is stable. To prevent cross-device scope creep, the cross-platform experience will be intentionally limited to thin, state-synchronized interactions for capture, monitoring, and approval, with more advanced UI features deferred beyond the MVP. Finally, to control domain breadth, the prototype will be validated on only three representative workflows, while the architecture will emphasize adapter-based extensibility so additional domains can be added without changing core system logic.